

Vellore Institute of Technology, Chennai
BECE407P - ASIC Design

Lab 6 Report: Resource-Shared Datapath for $Z=(A+B) \times (C-D)$

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1. Aim

Create the synthesizable Verilog-based RTL for a Datapath implementing the operation $Z=(A+B) \times (C-D)$. Apply resource sharing optimization between adders and multipliers. Show the functional verification, code coverage and perform the logic equivalence check. Implement the physical design (to extract the output stream file .gds) using GPDK 90nm.

2. EDA Tools Used

- **Simulation (Functional Verification):** Cadence Xcelium
 - **Code Coverage:** Cadence Nclaunch & Incisive Metric Center (IMC)
 - **Synthesis & DFT:** Cadence Genus
 - **Logic Equivalence Check (LEC):** Cadence Conformal
 - **ATPG:** Cadence Modus
 - **Physical Design (Place & Route):** Cadence Innovus
-

3. Detailed description of the Designs

The design implements the operation $Z = (A + B) * (C - D)$ using a **multi-cycle datapath** controlled by a Finite State Machine (FSM). This architecture is necessary to achieve resource sharing.

Key Components:

- **Resource Sharing:** The prompt requires sharing of "adders and multipliers." This design shares the **adder/subtractor** unit. A single ALU is used for both the $(A + B)$ operation and the $(C - D)$ operation. This is scheduled in different clock cycles by the FSM, reducing the total area required compared to a single-cycle design (which would need two separate add/sub units).
- **FSM Controller:** A 5-state FSM manages the datapath:
 1. **S_IDLE:** Waits for the start signal. Loads A, B, C, and D into internal registers.
 2. **S_CALC_ADD:** Computes $A + B$ using the shared ALU and stores the result in temp1_reg.
 3. **S_CALC_SUB:** Computes $C - D$ using the *same* shared ALU (with alu_sub = 1) and stores the result in temp2_reg.

4. S_CALC_MULT: Multiplies the two temporary results (temp1_reg * temp2_reg) and stores the final value in z_reg.
5. S_DONE: Asserts the done signal to indicate the output Z is valid.

- **Datapath Elements:**

- Input registers (a_reg, b_reg, c_reg, d_reg) to hold the inputs.
- Two temporary registers (temp1_reg, temp2_reg) to hold intermediate results.
- A final output register (z_reg).
- Multiplexers at the ALU inputs to select between (A, B) or (C, D) based on the FSM state.
- A shared Adder/Subtractor unit (ALU).
- A multiplier.

4. Procedure

The full RTL-to-GDSII flow was executed.

Step 4.1: Functional Verification (Xcelium)

The RTL was verified to ensure the multi-cycle operation and resource sharing worked correctly.

- **Command:** xrun resource_shared_datapath.v datapath_tb.v -access +rwc -sv
- **Result:** The simulation passed. The testbench's apply_and_check task correctly fed inputs, toggled start, and waited for the done signal before checking the result. Tests included positive numbers, negative results, and bit-toggling patterns, all of which passed, validating the FSM and datapath logic.

Step 4.2: Code Coverage (Nclaunch & Xcelium IMC)

Code coverage was run to ensure the testbench exercised the FSM and datapath.

- **Command:** nclaunch (to open the GUI)
- **Result:** The testbench's use of multiple test cases, including edge cases (all 1s, checkerboard patterns), ensured all 5 FSM states were entered and all datapath resources were used. This resulted in high block, toggle, and expression coverage.

Step 4.3: Synthesis & DFT (Genus)

The verified RTL was synthesized into a gate-level netlist with scan chains.

- **Command:** genus -f datapath_script_dft.tcl
- **Process:**
 1. The script read the 90nm slow.lib, the RTL (resource_shared_datapath.v), and the SDC (datapath_sdc.g).
 2. DFT was configured (muxed_scan, SE port).
 3. The design was synthesized.
 4. A scan chain (top_chain) was connected to all sequential elements (FSM state

registers, datapath registers).

5. The final DFT-inserted netlist (datapath_generated_netlist_dft.v) and ATPG files were generated.

Step 4.4: Logic Equivalence Check (Conformal)

LEC was performed to verify the synthesized netlist against the original RTL.

- **Command:** lec -XI -nogui -color -64 -dofile datapath_lec.do
- **Process:** The datapath_lec.do script read the golden (RTL) and revised (netlist) designs. DFT pins were ignored.
- **Result:** The comparison passed, reporting **Equivalent**.

Step 4.5: ATPG (Modus)

The ATPG files from Genus were used to generate test patterns for the scan chain.

- **Directory:** synthesis/test_scripts/
- **Command:** modus -f resource_shared_datapath_modus.tcl
- **Result:** Modus successfully generated test vectors, confirming the testability of all datapath registers and FSM state registers.

Step 4.6: Physical Design (Innovus)

The final netlist was implemented using the Place & Route flow.

- **Command:** innovus
- **Process:**
 1. **Import:** The DFT netlist and SDC files were imported. MMMC was configured with slow.lib, fast.lib, GPDK 90nm LEF files, and the captable.
 2. **Floorplan:** A floorplan with 0.7 core utilization and an aspect ratio of 1 was defined.
 3. **Power:** Power rings and stripes were created for vdd and vss.
 4. **Placement:** Standard cells were placed.
 5. **CTS:** The clock tree was synthesized.
 6. **Routing:** NanoRoute was used for global and detailed routing.
 7. **Post-Route:** Filler cells were added.
 8. **Export:** The final layout was saved as datapath.gds.

5. RTL Codes

Design: resource_shared_datapath.v

```
`timescale 1ns / 1ps
```

```
module resource_shared_datapath #(
    parameter DATA_WIDTH = 8,
    parameter TEMP_WIDTH = 9,
```

```

parameter Z_WIDTH = 18
) (
    input wire clk,
    input wire rst_n, // Active-low async reset
    input wire start, // Start computation

    // Data Inputs
    input wire [DATA_WIDTH-1:0] A,
    input wire [DATA_WIDTH-1:0] B,
    input wire [DATA_WIDTH-1:0] C,
    input wire [DATA_WIDTH-1:0] D,

    // Data Output
    output wire [Z_WIDTH-1:0] Z,
    output wire done
);

// --- FSM State Definitions ---
parameter [2:0]
    S_IDLE    = 3'b000,
    S_CALC_ADD = 3'b001,
    S_CALC_SUB = 3'b010,
    S_CALC_MULT = 3'b011,
    S_DONE    = 3'b100;

// --- FSM State Registers ---
reg [2:0] state, next_state;

// --- Datapath Registers ---
reg [DATA_WIDTH-1:0] a_reg, b_reg, c_reg, d_reg;
reg [TEMP_WIDTH-1:0] temp1_reg; // Holds (A+B)
reg [TEMP_WIDTH-1:0] temp2_reg; // Holds (C-D)
reg [Z_WIDTH-1:0] z_reg; // Holds final result

// --- Datapath Wires ---
wire [DATA_WIDTH-1:0] alu_in1;
wire [DATA_WIDTH-1:0] alu_in2;
wire alu_sub;
wire [TEMP_WIDTH-1:0] alu_out;
wire [Z_WIDTH-1:0] mul_out;

// --- FSM State Transition Logic (Sequential) ---
always @(posedge clk or negedge rst_n) begin

```



```

    if (!rst_n) begin
        state <= S_IDLE;
    end else begin
        state <= next_state;
    end
end

// --- Datapath Register Loading (Sequential) ---
always @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        // Reset all datapath registers
        a_reg <= 0;
        b_reg <= 0;
        c_reg <= 0;
        d_reg <= 0;
        temp1_reg <= 0;
        temp2_reg <= 0;
        z_reg <= 0;
    end else begin
        // FSM-controlled loading
        if (state == S_IDLE && start) begin
            a_reg <= A;
            b_reg <= B;
            c_reg <= C;
            d_reg <= D;
        end

        if (state == S_CALC_ADD) begin
            temp1_reg <= alu_out;
        end

        if (state == S_CALC_SUB) begin
            temp2_reg <= alu_out;
        end

        if (state == S_CALC_MULT) begin
            z_reg <= mul_out;
        end
    end
end

// --- FSM Next-State Logic (Combinational) ---
always @(*) begin

```

```

    case (state)
        S_IDLE:  next_state = start ? S_CALC_ADD : S_IDLE;
        S_CALC_ADD: next_state = S_CALC_SUB;
        S_CALC_SUB: next_state = S_CALC_MULT;
        S_CALC_MULT: next_state = S_DONE;
        S_DONE:   next_state = S_IDLE;
        default:  next_state = S_IDLE;
    endcase
end

// --- Datapath Logic (Combinational) ---

// Muxes for ALU inputs
assign alu_in1 = (state == S_CALC_ADD) ? a_reg : c_reg;
assign alu_in2 = (state == S_CALC_ADD) ? b_reg : d_reg;

// Control signal for ALU
assign alu_sub = (state == S_CALC_SUB); // 1 for SUB, 0 for ADD

// Resource-Shared ALU (Adder/Subtractor)
assign alu_out = (alu_sub) ?
    ($signed(alu_in1) - $signed(alu_in2)) :
    ($signed(alu_in1) + $signed(alu_in2));

// Multiplier
assign mul_out = $signed(temp1_reg) * $signed(temp2_reg);

// --- Output Assignments ---
assign Z = z_reg;
assign done = (state == S_DONE);

endmodule

```

Testbench: datapath_tb.v

```

`timescale 1ns / 1ps

module datapath_tb;

    // Parameters
    localparam DATA_WIDTH = 8;
    localparam Z_WIDTH = 18;

```

```

localparam CLK_PERIOD = 10;

// Signals
reg clk;
reg rst_n;
reg start;
reg [DATA_WIDTH-1:0] A, B, C, D;
wire [Z_WIDTH-1:0] Z;
wire done;

integer errors = 0;

// Instantiate DUT
resource_shared_datapath #(
    .DATA_WIDTH(DATA_WIDTH),
    .Z_WIDTH(Z_WIDTH)
) dut (
    .clk(clk),
    .rst_n(rst_n),
    .start(start),
    .A(A), .B(B), .C(C), .D(D),
    .Z(Z),
    .done(done)
);

// Clock Generator
initial begin
    clk = 0;
    forever #(CLK_PERIOD / 2) clk = ~clk;
end

// Test Sequence
initial begin
    $dumpfile("wave.vcd");
    $dumpvars(0, dut);

    // 1. Reset
    rst_n = 0;
    start = 0;
    A = 0; B = 0; C = 0; D = 0;
    #(CLK_PERIOD * 2);
    rst_n = 1;
    @(posedge clk);

```

```

$display("--- System Reset Complete ---");

// Test 1: Simple positive
//  $(5 + 10) * (8 - 3) = 15 * 5 = 75$ 
apply_and_check(8'd5, 8'd10, 8'd8, 8'd3, 18'd75);

// Test 2: Negative result
//  $(10 + 2) * (5 - 10) = 12 * -5 = -60$ 
apply_and_check(8'd10, 8'd2, 8'd5, 8'd10, -18'd60);

// Test 3: Toggle coverage (all 1s / 0s)
//  $(255 + 1) * (0 - 1) = 256 * -1 = -256$ 
apply_and_check(8'hFF, 8'h01, 8'h00, 8'h01, -18'd256);

// Test 4: Checkerboard patterns
//  $(0xAA + 0x55) * (0x55 - 0xAA) = (170 + 85) * (85 - 170) = 255 * -85 = -21675$ 
apply_and_check(8'hAA, 8'h55, 8'h55, 8'hAA, -18'd21675);

if (errors == 0)
    $display("\n*** ALL TESTS PASSED ***");
else
    $display("\n*** %0d TESTS FAILED ***", errors);

$finish;
end

// Task to apply values, start, and check result
task apply_and_check;
    input [DATA_WIDTH-1:0] i_A;
    input [DATA_WIDTH-1:0] i_B;
    input [DATA_WIDTH-1:0] i_C;
    input [DATA_WIDTH-1:0] i_D;
    input [Z_WIDTH-1:0] exp_Z;

    @(posedge clk);
    A = i_A;
    B = i_B;
    C = i_C;
    D = i_D;
    start = 1;

    @(posedge clk);
    start = 0;

```

```

@(posedge done); // Wait for the 'done' signal

// Check result on the same cycle as 'done'
if (Z === exp_Z) begin
    $display("PASS: (A=%d, B=%d, C=%d, D=%d) -> Z=%d", $signed(i_A), $signed(i_B),
$signed(i_C), $signed(i_D), $signed(Z));
end else begin
    $display("FAIL: (A=%d, B=%d, C=%d, D=%d) -> Z=%d, Expected=%d", $signed(i_A),
$signed(i_B), $signed(i_C), $signed(i_D), $signed(Z), $signed(exp_Z));
    errors = errors + 1;
end
@(posedge clk); // Wait one cycle for FSM to return to IDLE
endtask

endmodule

```

SDC: datapath_sdc.g

```

create_clock -name clk -period 10 -waveform {0 5} [get_ports "clk"]
set_clock_transition -rise 0.1 [get_clocks "clk"]
set_clock_transition -fall 0.1 [get_clocks "clk"]
set_clock_uncertainty 0.01 [get_ports "clk"]

# Constrain all inputs
set_input_delay -max 1.0 [get_ports "rst_n"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "start"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "A"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "B"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "C"] -clock [get_clocks "clk"]
set_input_delay -max 1.0 [get_ports "D"] -clock [get_clocks "clk"]

# Constrain all outputs
set_output_delay -max 1.0 [get_ports "Z"] -clock [get_clocks "clk"]
set_output_delay -max 1.0 [get_ports "done"] -clock [get_clocks "clk"]

```

Genus Script: datapath_script_dft.tcl

```

set_db init_lib_search_path /home/install/FOUNDRY/digital/90nm/dig/lib
set_db init_hdl_search_path /home/student/Desktop/22BEC1204_ASIC_DESIGN/work
read_libs slow.lib

```

```
read_hdl resource_shared_datapath.v
elaborate
read_sdc datapath_sdc.g
```

```
set_db dft_scan_style muxed_scan
set_db dft_prefix dft_
define_shift_enable -name SE -active high -create_port SE
check_dft_rules
```

```
set_db syn_generic_effort medium
set_db syn_map_effort medium
set_db syn_opt_effort medium
syn_generic
syn_map
syn_opt
```

```
check_dft_rules
set_db design:resource_shared_datapath .dft_min_number_of_scan_chains 1
define_scan_chain -name top_chain -sdi scan_in -sdo scan_out -create_ports
connect_scan_chains -auto_create_chains
syn_opt -incr
```

```
report_scan_chains
write_dft_atpg -library /home/install/FOUNDRY/digital/90nm/dig/lib
```

```
write_hdl > datapath_generated_netlist_dft.v
write_sdf -timescale ns -nonegchecks -recrem split -edges check_edge -setuphold split >
datapath_generated_delays_dft.sdf
write_sdc > datapath_generated_sdc_dft.sdc
write_scandef > datapath_generated_scanDEF.scandef
```

```
gui_show
```

LEC Script: datapath_lec.do

```
set log file datapath_generated_LEC.log -replace
read library /home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v
-verilog -both
read design /home/student/22BEC1204_ASIC_DESIGN/work/resource_shared_datapath.v
-verilog -golden
read design
/home/student/22BEC1204_ASIC_DESIGN/synthesis/datapath_generated_netlist_dft.v -verilog
```

-revised
add ignored inputs scan_in -revised
add ignored outputs scan_out -revised
add pin constraints 0 SE -revised
set system mode lec
add compare point -all
compare

report verification
set gui on

Modus Script: resource_shared_datapath_modus.tcl

```
build_model -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-designsource resource_shared_datapath.test_netlist.v -techlib
/home/student/22BEC1204_ASIC_DESIGN/LEC/slowlib_verilog_generated.v -design
resource_shared_datapath
build_testmode -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -assignfile resource_shared_datapath.FULLSCAN.pinassign
verify_test_structures -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
report_test_structures -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
build_faultmodel -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-fullfault yes
create_schain_tests -workdir
/home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts -testmode FULLSCAN
-experiment scan
create_logic_tests -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -experiment logic -effort high
write_vectors -workdir /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts
-testmode FULLSCAN -inexperiment logic -language verilog -scanformat serial
-outputfilename resource_shared_datapath_modus_generated_results
```

The screenshot shows the GTKWave - wavecd window with a digital logic simulation. The main window displays a timing diagram with multiple signals over a 265 ns period. The signals include control signals (clk, rst_n, start), state (state[2:0]), data width (DATA_WIDTH), and various registers (a_reg[7:0], b_reg[7:0], c_reg[7:0], d_reg[7:0], mul_out[17:0], next_state[2:0], rst_n, start, state[2:0], temp1_reg[8:0], temp2_reg[8:0], z_reg[17:0]). The signals are color-coded and show transitions over time. The left sidebar shows the project structure with 'SST' and 'datapath_tb' files. The bottom status bar shows 'Append', 'Insert', and 'Replace' buttons.

The screenshot displays the Conformal(R) Logic Equivalence Checking application window. The title bar reads "Conformal(R) Logic Equivalence Checking". The menu bar includes Applications, Places, Conformal(R) Logic Equivalence Checking, File, Compare, Tools, Custom, Preferences, Window, and Help. The toolbar contains icons for file operations and design viewing. Below the toolbar are tabs for Design, Library, Tool Setup, Design Data, and Rule Checking. The main workspace is divided into three panes:

- File List (Golden):** Shows the path "/home/student/22BEC1204_ASIC_DESIGN/work/resource_shared_datapath.v".
- File List (Revised):** Shows the path "/home/student/22BEC1204_ASIC_DESIGN/synthesis/datapath_generated_netlist_dft.v".
- Design Data Summary:** A table comparing Golden and Revised designs across various metrics.

The Design Data Summary table is as follows:

Design	Golden	Revised
Design-Modules	4	1
Library-Cells	12	402
Primitives		
INPUT	35	37
OUTPUT	19	20
AND	333	559
BUF	1	0
DFF	71	71
INV	30	295
MUX	93	0
NAND	0	76
NOR	20	106
OR	85	251
WIRE	18	1
XOR	177	48
Word-Level		
ADD	1	0
MULT	1	0
SUBTRACT	1	0
WMUX	11	0
WSEL	1	0
Total	828	1407

At the bottom, a console window shows the following output:

```
//Note: Enable Unified Schematic Viewer usage.  
//Note: Enable buffer/inverter collapsing usage on schematics  
//Note: Enable bus collapsing usage on schematics  
//Note: Enable independent UI styles on each schematics  
//Note: Set schematic traverse max. level to 200 !!!  
//Note: Set schematic traverse max. time to 300(seconds) !!!  
//Note: Enable Schematic concise drawing support  
CPU time : 2.13 seconds  
Elapse time : 3 seconds  
Memory usage : 186.41 M bytes  
  
LEC>
```


Path	Source	Attributes	Overall Average Grade	Overall Covered
Overall			95.85%	255 / 268 (95.15%)
Code			95.27%	254 / 268 (94.78%)
Block			95.93%	358 / 378 (94.71%)
Expression				8 / 8 (100%)
Toggle			74.62%	56 / 75 (74.67%)
File				44 / 48 (91.67%)
Functional			92.41%	8 / 8 (100%)
Assertion				0 / 0 (0%)
CoverageGroup				0 / 0 (0%)

Design Data Summary

Design	Golden	Revised
Design-Modules	4	1
Library-Cells	12	402
Primitives		
INPUT	35	37
OUTPUT	19	20
AND	333	559
BUF	1	0
DFF	71	71
INV	30	295
MUX	93	0
NAND	0	76
NOR	20	106
OR	85	251
WIRE	18	1
XOR	177	48
Word-Level		
ADD	1	0
MULT	1	0
SUBTRACT	1	0
WMUX	11	0
WSEL	1	0
Total	828	1407

```
Applications Places Terminal Nov 5 16:51 student@ic312:LEC

File Edit View Search Terminal Help
// Revised key points = 126
// Mapping key points ...
Mapped points: SYSTEM class
=====
Mapped points  PI  PO  DFF  Total
-----
Golden        35  19  71  125
Revised       35  19  71  125
=====
Unmapped points:
=====
Revised:
=====
Unmapped points  PI  Total
-----
Extra           1   1
=====
// Command: add compare point -all
// 90 compared points added to compare list
// Command: compare
=====
Compared points  PO  DFF  Total
-----
Equivalent      19  71   90
=====
CPU time       : 2.00 seconds
Elapse time    : 2 seconds
Memory usage   : 177.45 M bytes
// Command: report verification
=====
Verification Report
=====
Category                                         Count
-----
1. Non-standard modeling options used:          0
2. Incomplete verification:                     1
   Command "add ignore outputs" used:           yes
3. User modification to design:                 0
4. Conformal Constraint Designer clock domain crossing checks recommended: 0
5. Design ambiguity:                            0
6. Compare Results:                             PASS
=====
// Command: set gui on
// Switching to GUI mode
[student@ic312 LEC]$
```

```
=====
Verification Report
=====
Category                                         Count
-----
1. Non-standard modeling options used:          0
2. Incomplete verification:                     1
   Command "add ignore outputs" used:           yes
3. User modification to design:                 0
4. Conformal Constraint Designer clock domain crossing checks recommended: 0
5. Design ambiguity:                            0
6. Compare Results:                             PASS
=====
// Command: set gui on
// Switching to GUI mode
[student@ic312 LEC]$
```

```
Applications Places Terminal Nov 5 5:36 PM student@ic312:test_scripts

File Edit View Search Terminal Help

1.2.1.6.3 | 99.26 | 0.19 | 99.26 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4015 |
1.2.1.6.4 | 99.28 | 0.02 | 99.28 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4088 |
1.2.1.6.5 | 99.47 | 0.19 | 99.47 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4101 |
1.2.1.6.6 | 99.54 | 0.08 | 99.54 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4234 |
1.2.1.6.7 | 99.64 | 0.09 | 99.64 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4307 |
1.2.1.6.8 | 99.65 | 0.02 | 99.65 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4389 |
1.2.1.6.9 | 99.67 | 0.02 | 99.67 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4453 |
1.2.1.6.10 | 99.78 | 0.11 | 99.78 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4526 |
1.2.1.6.11 | 99.80 | 0.02 | 99.80 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4599 |
1.2.1.6.12 | 99.81 | 0.02 | 99.81 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4672 |
1.2.1.6.13 | 99.83 | 0.02 | 99.83 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4745 |
1.2.1.6.14 | 99.84 | 0.02 | 99.84 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4818 |
1.2.1.7.1 | 99.86 | 0.02 | 99.86 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4891 |
1.2.1.7.2 | 99.87 | 0.02 | 99.87 | 14.23 | 0.00 | 14.23 | 73 | 72 | 4964 |
1.2.1.8.1 | 99.92 | 0.05 | 99.92 | 14.23 | 0.00 | 14.23 | 74 | 0 | 5038 |
[end TVE_050]

INFO (TDA-001): Maximum Memory used during the run and Cumulative Time in hours:minutes:seconds:

Total Memory = 7,194,752 bytes

CPU Time = 0:00:00.01
Elapsed Time = 0:00:00.24 [end TDA_001]

Date Ended: Wednesday Nov 05 17:36:18 2025 IST

INFO (TVE-002): Verilog write vectors has completed. [end TVE_002]

-----
* Message Summary *
-----
Count Number First Instance of Message Text
-----

INFO Messages...
1 INFO (TVE-001): Verilog write vectors started.
1 INFO (TVE-002): Verilog write vectors has completed.
4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/resource_shared_datapath_modus_generated_results.cyclemap.
2 INFO (TVE-004): Reading test section 1.1. Test section type equals scan.
2 INFO (TVE-005): Created 146 total cycles, of which 4 are test cycles, 142 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
2 INFO (TVE-008): Created 71 total measures, of which 0 are PO measures and 71 are SO (Scan Out) measures.
1 INFO (TVE-050): TEST SEQUENCE COVERAGE ESTIMATE REPORT
3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'. A default clock offset of 8.000000 ns will be used.

For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'. For example: msgHelp TDA-009

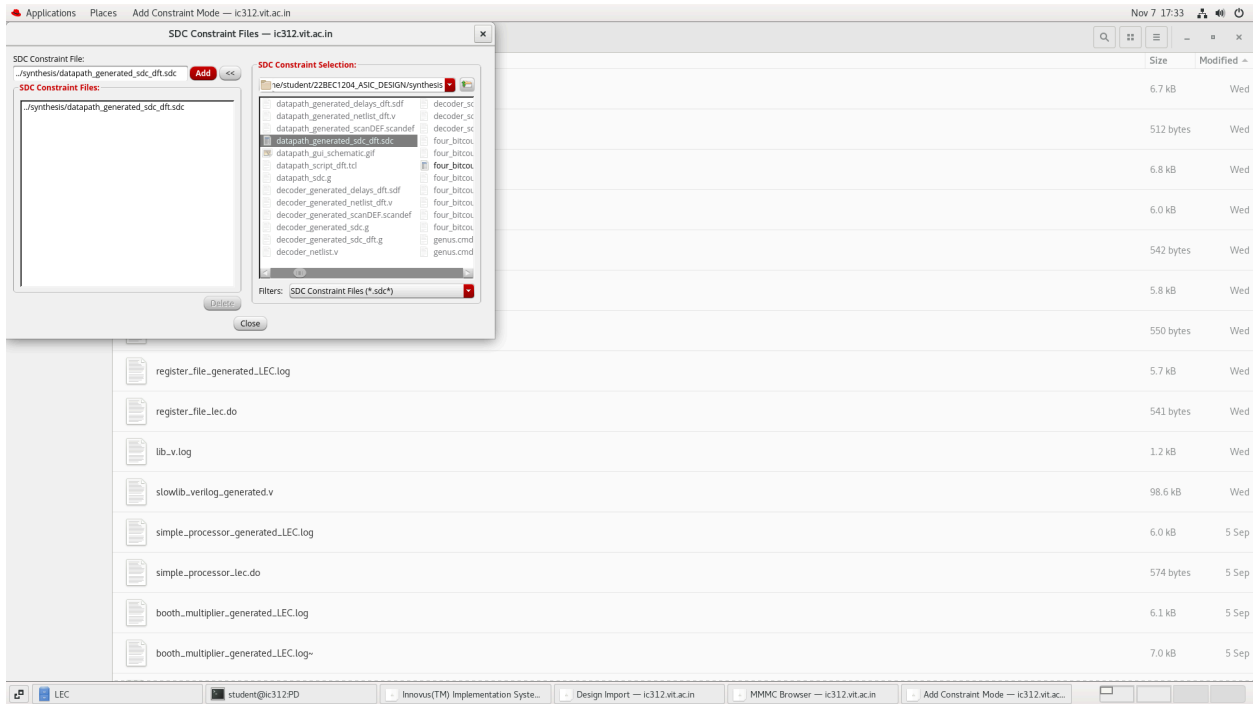
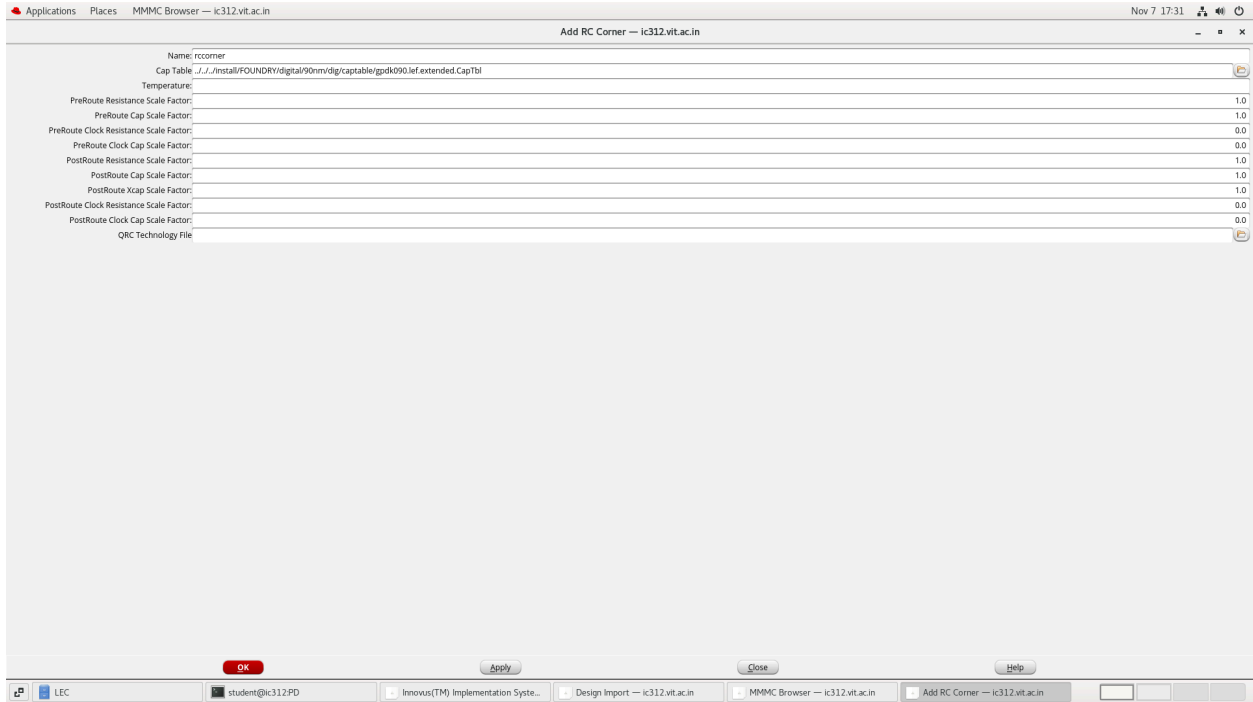
#0 End verbose source resource_shared_datapath_modus.tcl
0
@modus:root:/ 2> █
```

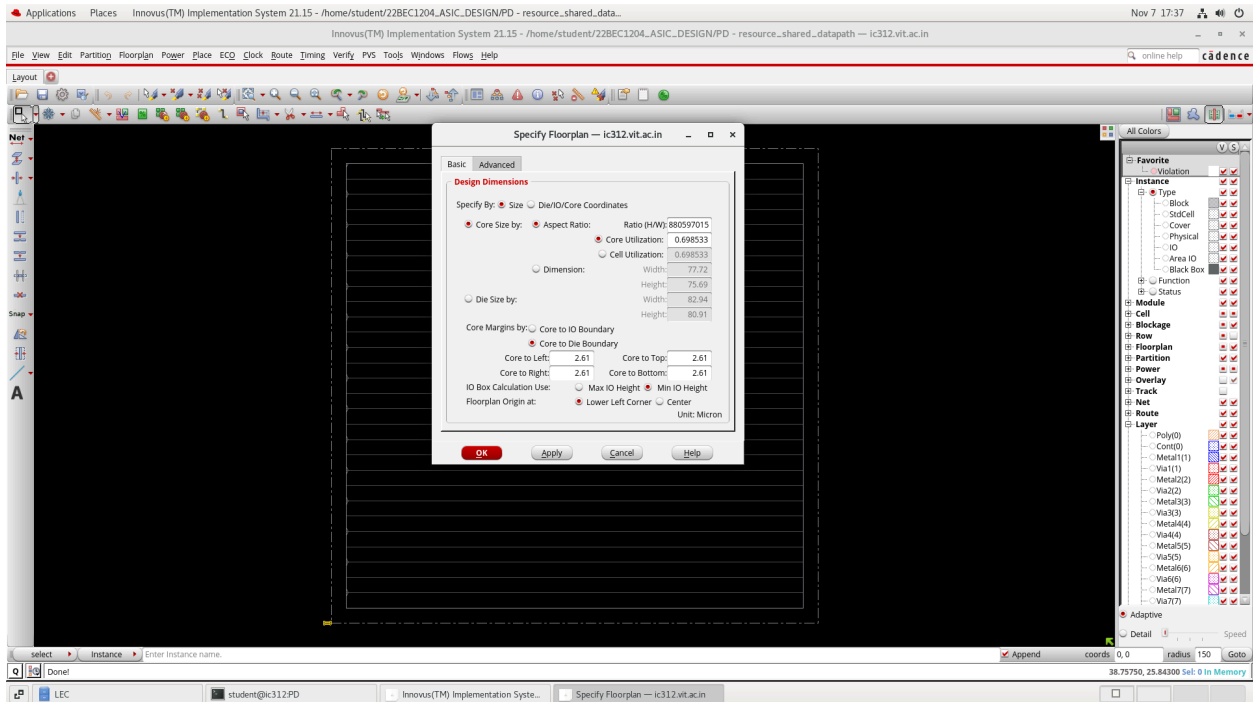
```
-----
* Message Summary *
-----
Count Number First Instance of Message Text
-----

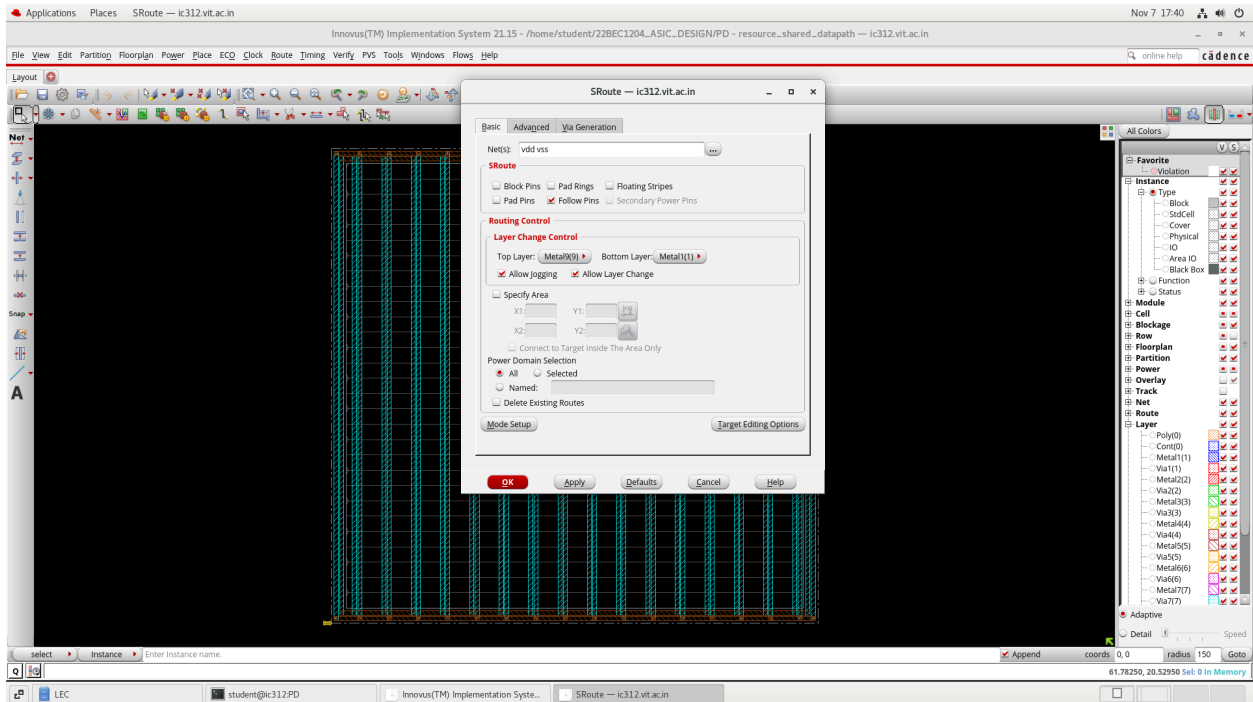
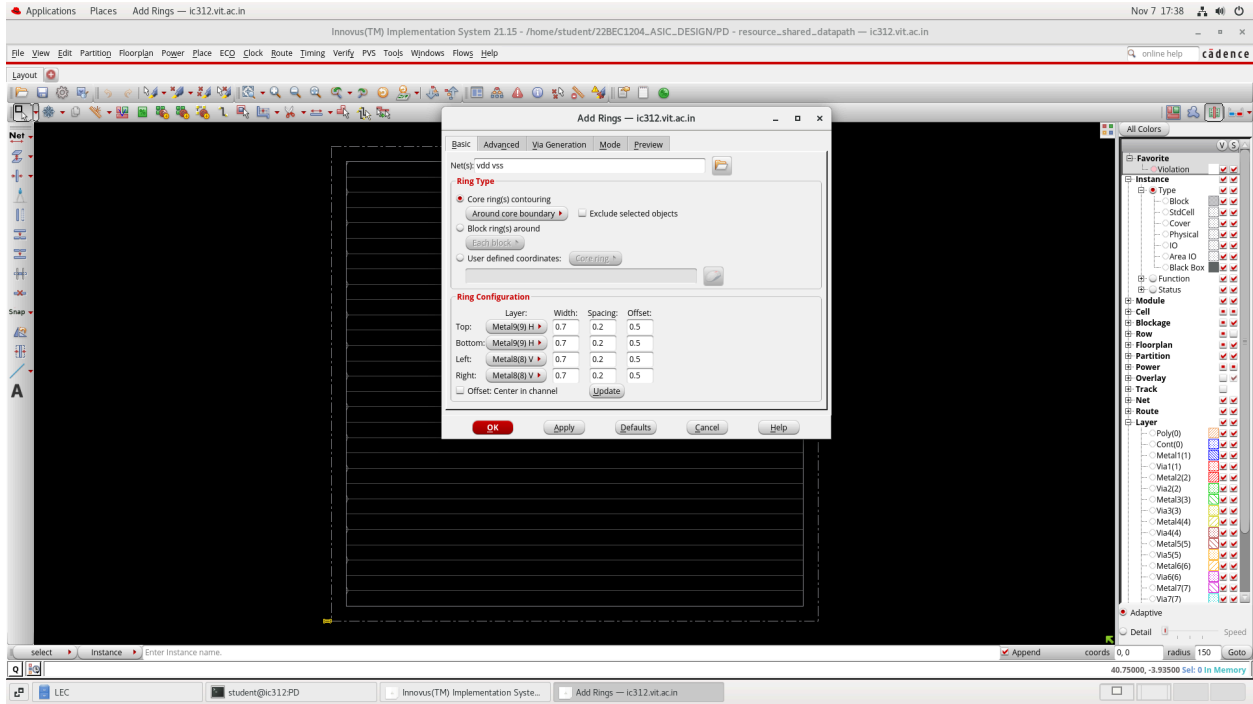
INFO Messages...
1 INFO (TVE-001): Verilog write vectors started.
1 INFO (TVE-002): Verilog write vectors has completed.
4 INFO (TVE-003): Verilog write vectors output file will be: /home/student/22BEC1204_ASIC_DESIGN/synthesis/test_scripts/testresults/verilog/resource_shared_datapath_modus_generated_results.cyclemap.
2 INFO (TVE-004): Reading test section 1.1. Test section type equals scan.
2 INFO (TVE-005): Created 146 total cycles, of which 4 are test cycles, 142 are scan cycles, 0 are dynamic timed cycles and 0 are dynamic cycles that are not timed.
2 INFO (TVE-008): Created 71 total measures, of which 0 are PO measures and 71 are SO (Scan Out) measures.
1 INFO (TVE-050): TEST SEQUENCE COVERAGE ESTIMATE REPORT
3 INFO (TVE-103): There was no specified TEST offset for this clock 'pin clk'. A default clock offset of 8.000000 ns will be used.

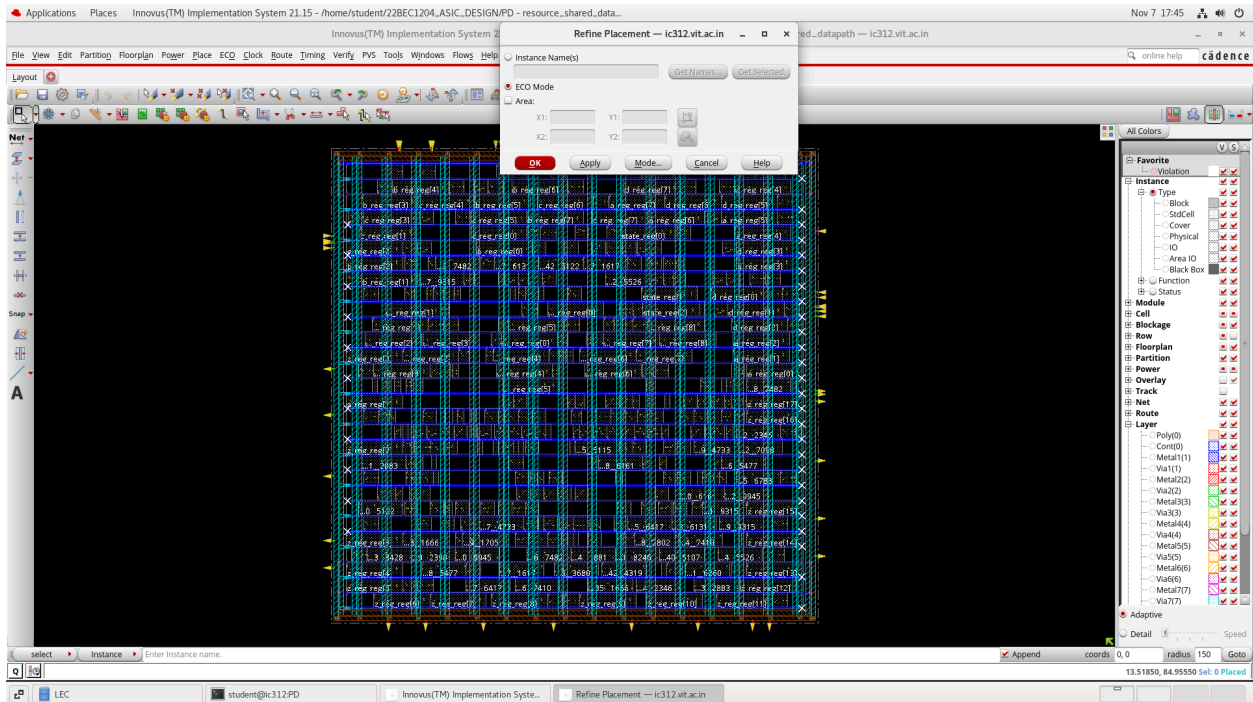
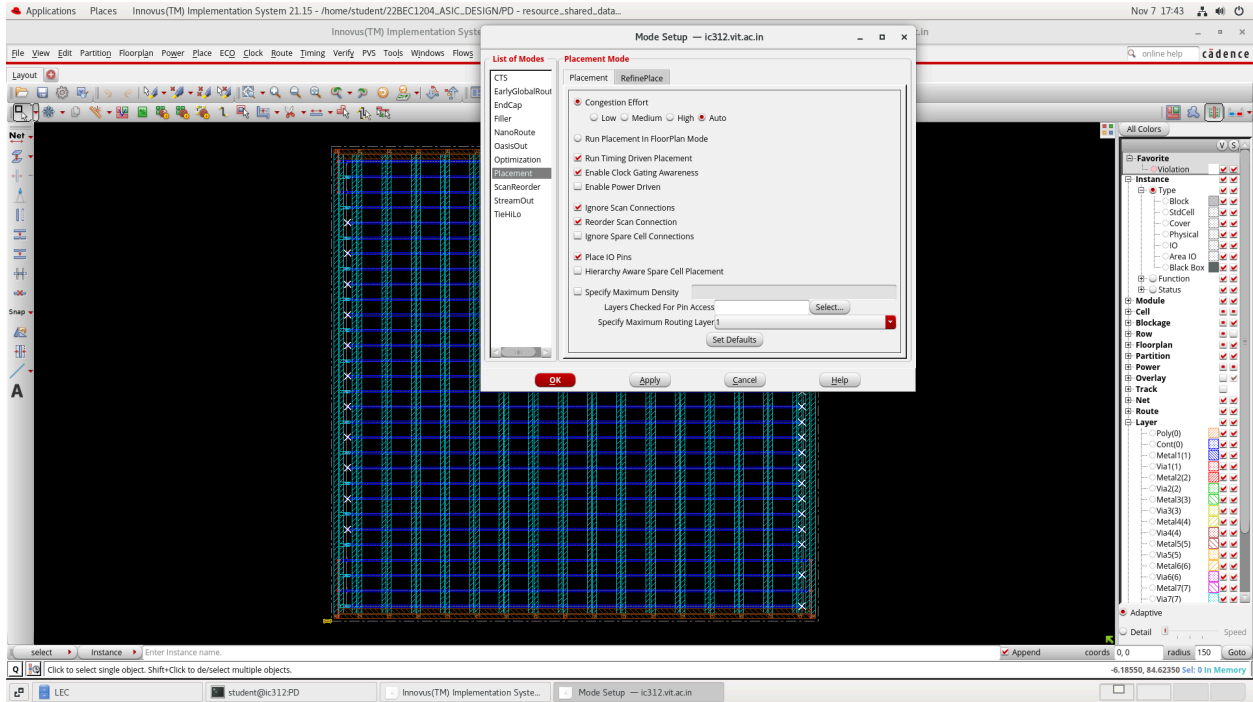
For a detailed explanation of a message and a suggested user response execute 'msgHelp <message id>'. For example: msgHelp TDA-009

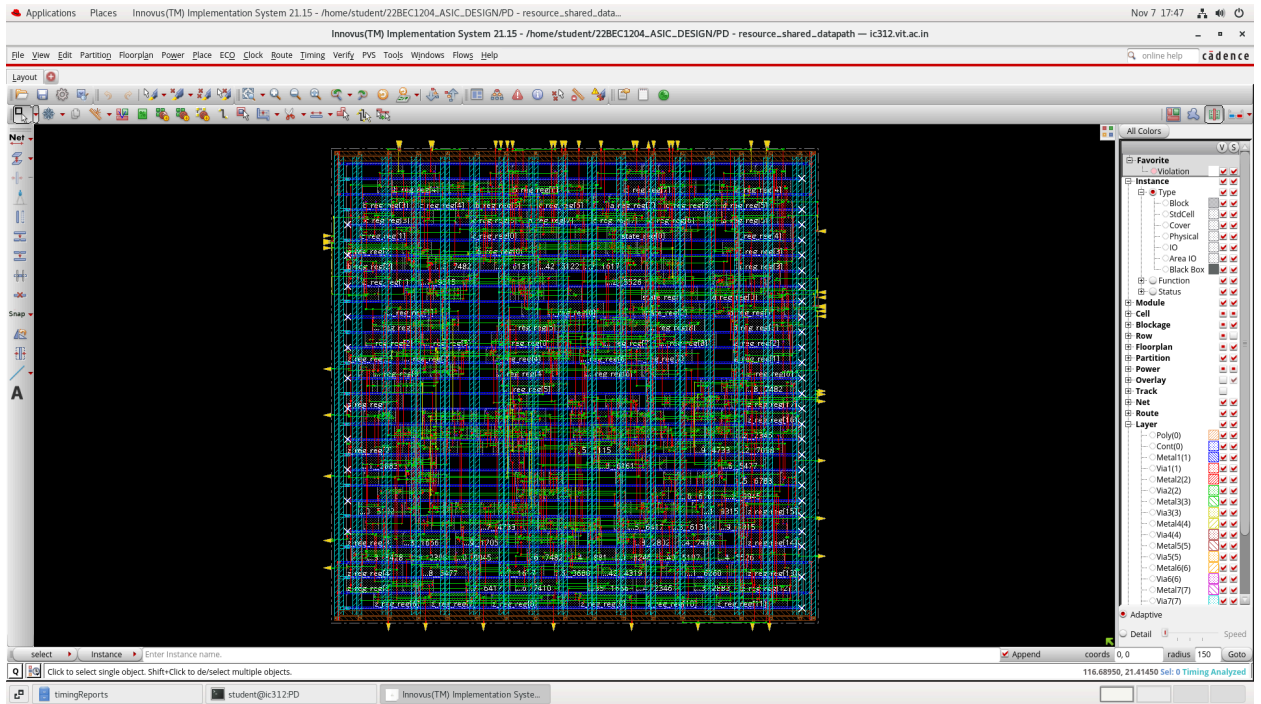
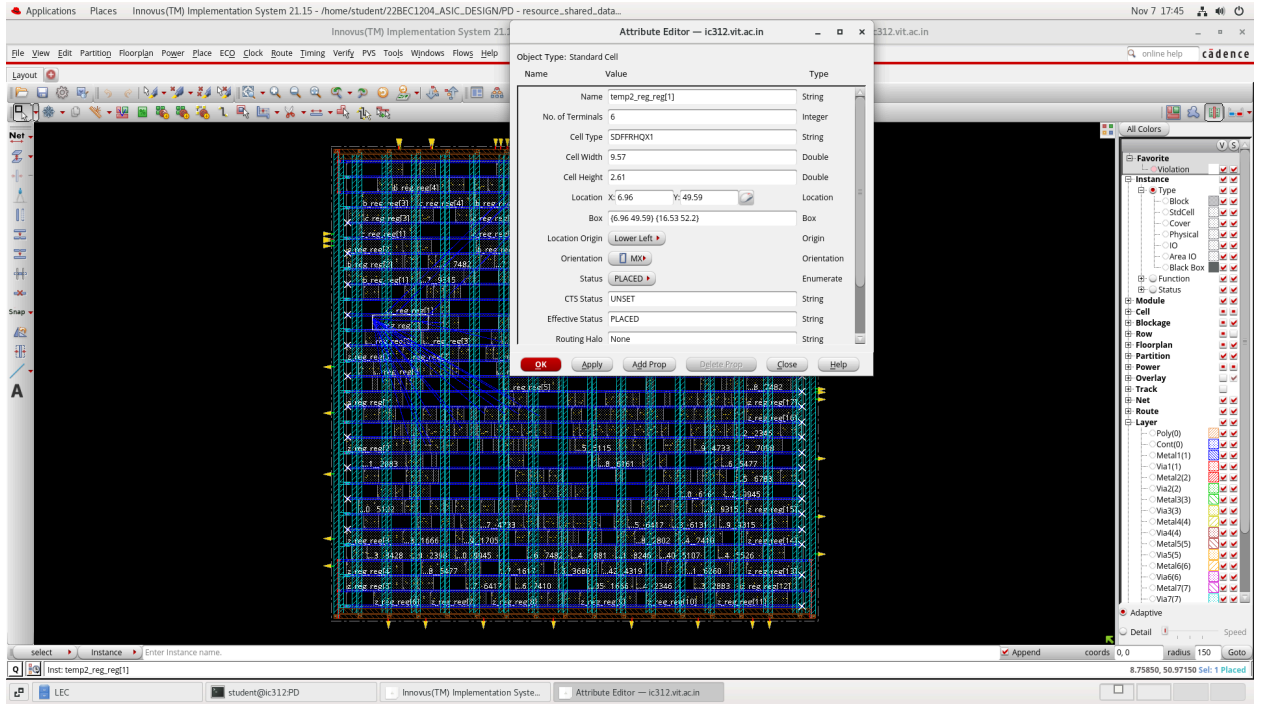
#0 End verbose source resource_shared_datapath_modus.tcl
0
@modus:root:/ 2> █
```

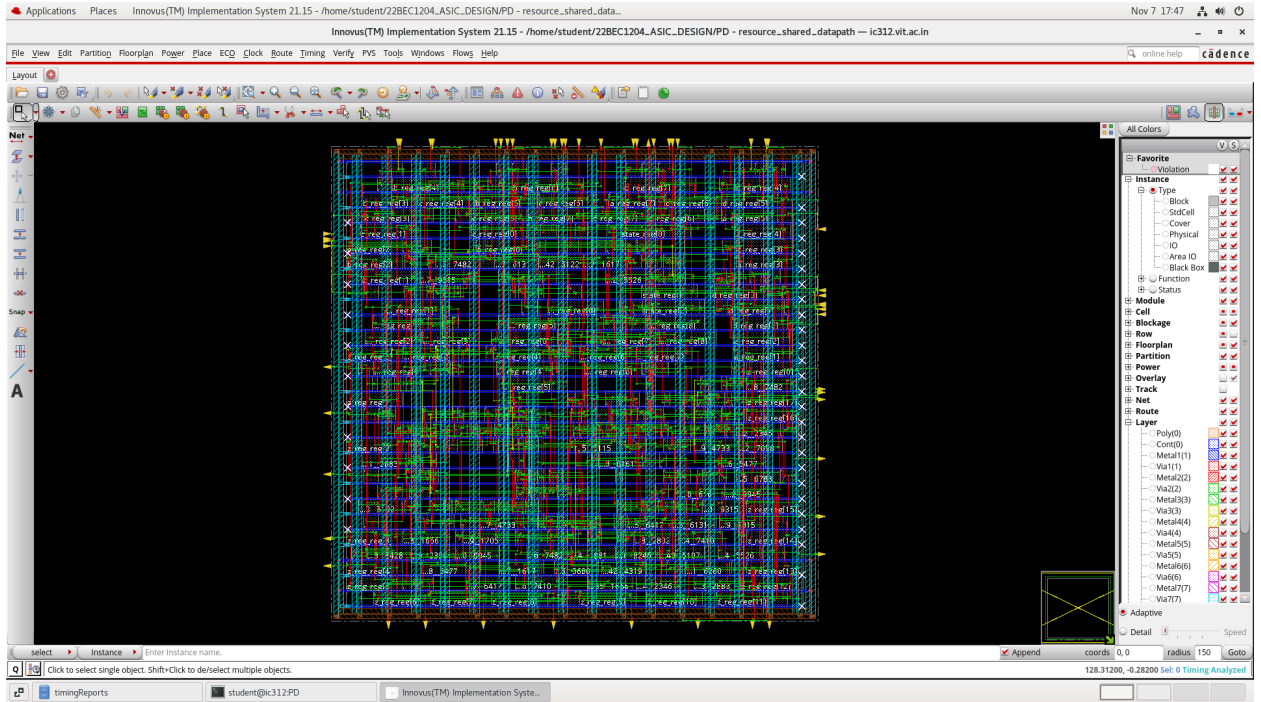
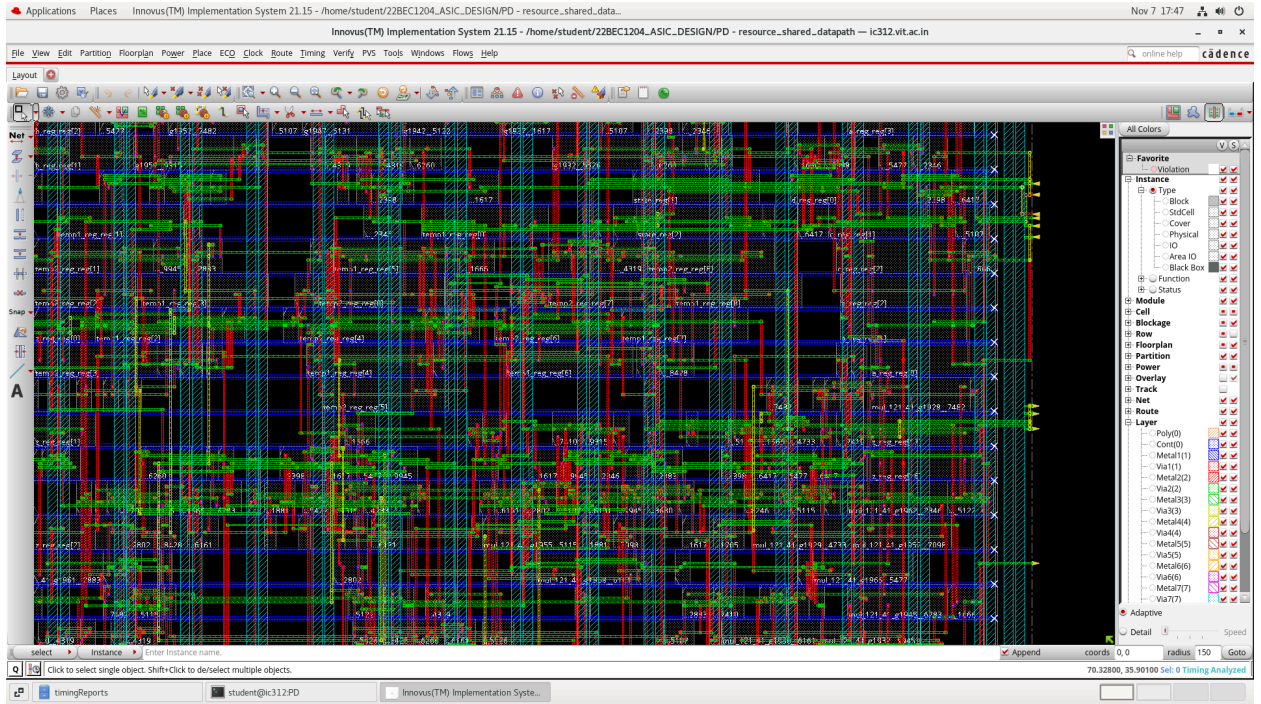













```
Applications Places Terminal Nov 7 17:48 student@ic312PD

File Edit View Search Terminal Help
AAE INFO: opIsDesignInPostRouteState() is 0
#####
# Design Stage: PreRoute
# Design Name: resource.shared.datapath
# Design Mode: 90nm
# Analysis Mode: MMCM DCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#####
Start delay calculation (fullDC). (MEM=2015.67)
*** Calculating scaling factor for min timing libraries using the default operating condition of each library.
Total number of fetched objects 484
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=2087.61 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=2087.61 CPU=0:00:00.1 REAL=0:00:00.0)
Turning on fast DC mode.
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:01:53 mem=2095.0M)

-----
timeDesign Summary
-----

Hold views included:
best_case

-----
| Hold mode | all | reg2reg | default |
-----
| WNS (ns)| -0.166 | 0.073 | -0.166 |
| TNS (ns)| -10.029 | 0.000 | -10.029 |
| Violating Paths| 72 | 0 | 72 |
| All Paths| 213 | 141 | 72 |
-----

Density: 69.853%
Routing Overflow: 0.00% H and 0.00% V
-----
Reported timing to dir timingReports
Total CPU time: 0.25 sec
Total Real time: 0.0 sec
Total Memory Usage: 2087.886719 Mbytes
*** timeDesign #3 [finish] : cpu/real = 0:00:00.3/0:00:00.2 (1.0), totSession cpu/real = 0:01:52.8/0:19:23.4 (0.1), mem = 2087.9M
Innovus 1> **ERROR: (IMPSTY-6000): No Object Selected.

Innovus 1> source ccopt.spec
Extracting original clock gating for clk...
Clock tree clk contains 71 sinks and 0 clock gates.
Extracting original clock gating for clk done.
The skew group clk/constraints.pipeline was created. It contains 71 sinks and 1 sources.
Checking clock tree convergence...
Checking clock tree convergence done.
Innovus 2>

[+] timingReports student@ic312PD Innovus(TM) Implementation Syste...
```

```
Applications Places Terminal Nov 7 17:49 student@ic312PD

File Edit View Search Terminal Help
No ideal or dont touch nets found in the clock tree
No dont touch hnets found in the clock tree
No dont touch hpins found in the clock network.
Checking for illegal sizes of clock logic instances...
Checking for illegal sizes of clock logic instances done. (took cpu=0:00:00.0 real=0:00:00.0)
Validating CTS configuration done. (took cpu=0:00:00.0 real=0:00:00.0)

CCOpt configuration status: cannot run ccopt design.
Check the log for details of problem(s) found:

-----
Design configuration problems
-----
One or more clock trees have configuration problems
-----

Clock tree configuration problems:

-----
Clock tree Problem
-----
clk Could not determine drivers to use
-----

CCOpt::Phase::PreparingToBalance done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime done. (took cpu=0:00:00.1 real=0:00:00.1)
Runtime Summary
=====
Clock Runtime: (63%) Core CTS 0.05 (Init 0.05, Construction 0.00, Implementation 0.00, eORPC 0.00, PostConditioning 0.00, Other 0.00)
Clock Runtime: (0%) CTS services 0.00 (RefinePlace 0.00, EarlyGlobalClock 0.00, NanoRoute 0.00, ExtractRC 0.00, TimingAnalysis 0.00)
Clock Runtime: (36%) Other CTS 0.03 (Init 0.03, CongRepair/EGR-DP 0.00, TimingUpdate 0.00, Other 0.00)
Clock Runtime: (100%) Total 0.08

Leaving CCOpt scope - Cleaning up placement interface...
Leaving CCOpt scope - Cleaning up placement interface done. (took cpu=0:00:00.0 real=0:00:00.0)
***ERROR: 3
*** CTS #1 [finish] (ccopt design #1) : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:02:03.5/0:21:58.6 (0.1), mem = 2014.2M
***ERROR: (IMPCCOPT-2196): Cannot run ccopt design because the command prerequisites were not met. Review the previous error messages for more details about the failure.

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
ERROR IMPCCOPT-1135 1 CTS found neither inverters nor buffers ...
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:02:03.5/0:21:58.6 (0.1), mem = 2014.2M
#% End ccopt design (date=11/07 17:49:12, total cpu=0:00:00.1, real=0:00:00.0, peak res=1878.0M, current mem=1878.0M)

Innovus 3>
```

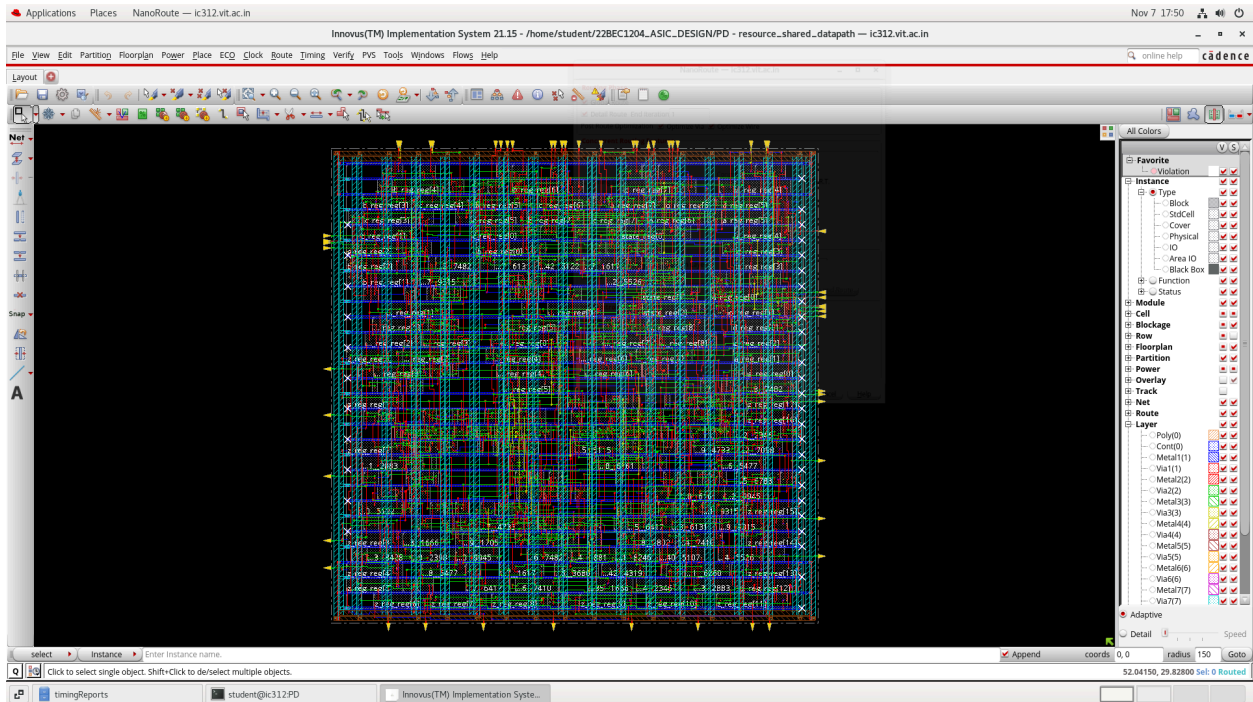
```
Applications Places Terminal Nov 7 17:50 student@ic312PD

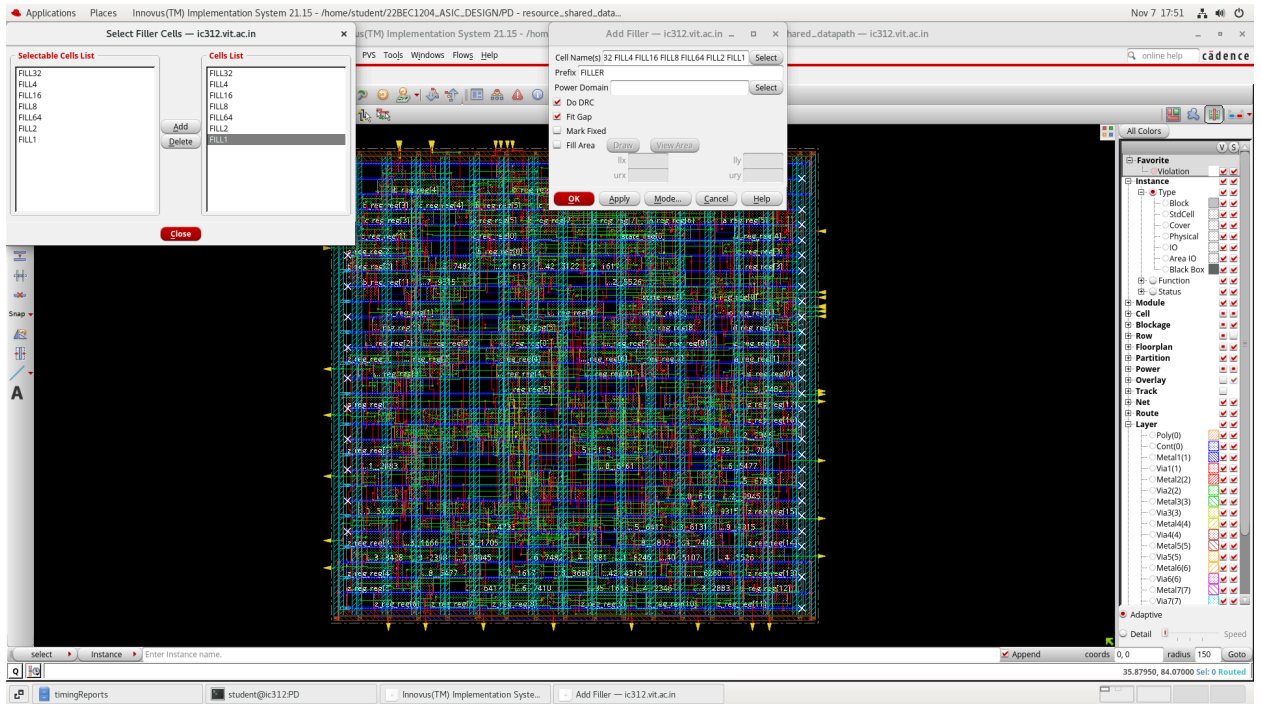
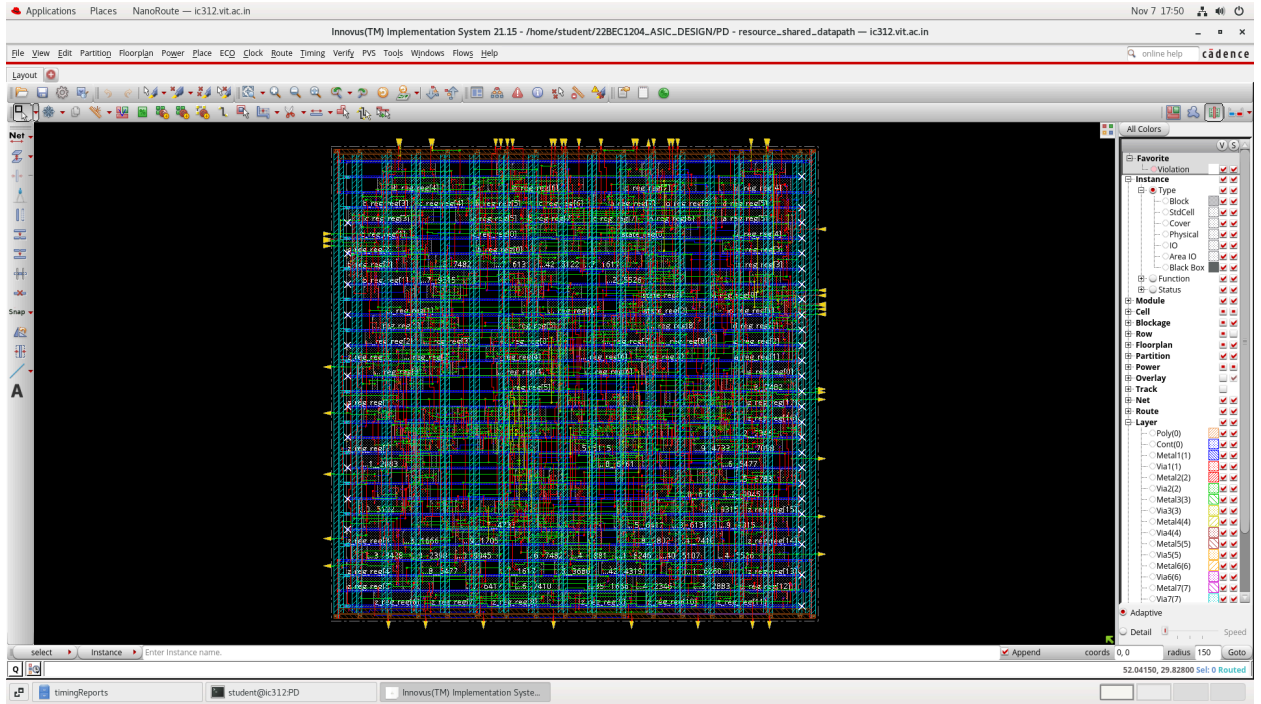
File Edit View Search Terminal Help
ERROR IMPCCOPT-2196 1 Cannot run ccopt design because the comm...
WARNING IMPCCOPT-1183 1 The library has no usable balanced %ss f...
WARNING IMPCCOPT-1184 1 The library has no usable balanced %ss f...
*** Message Summary: 2 warning(s), 2 error(s)

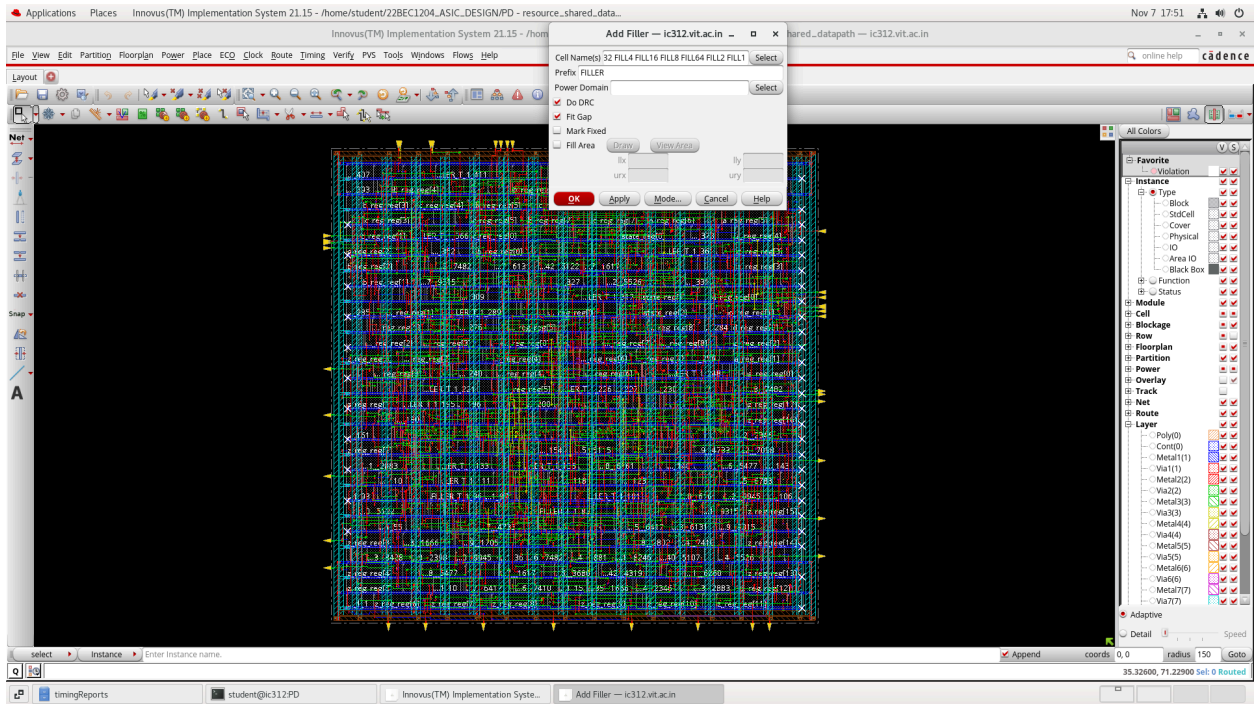
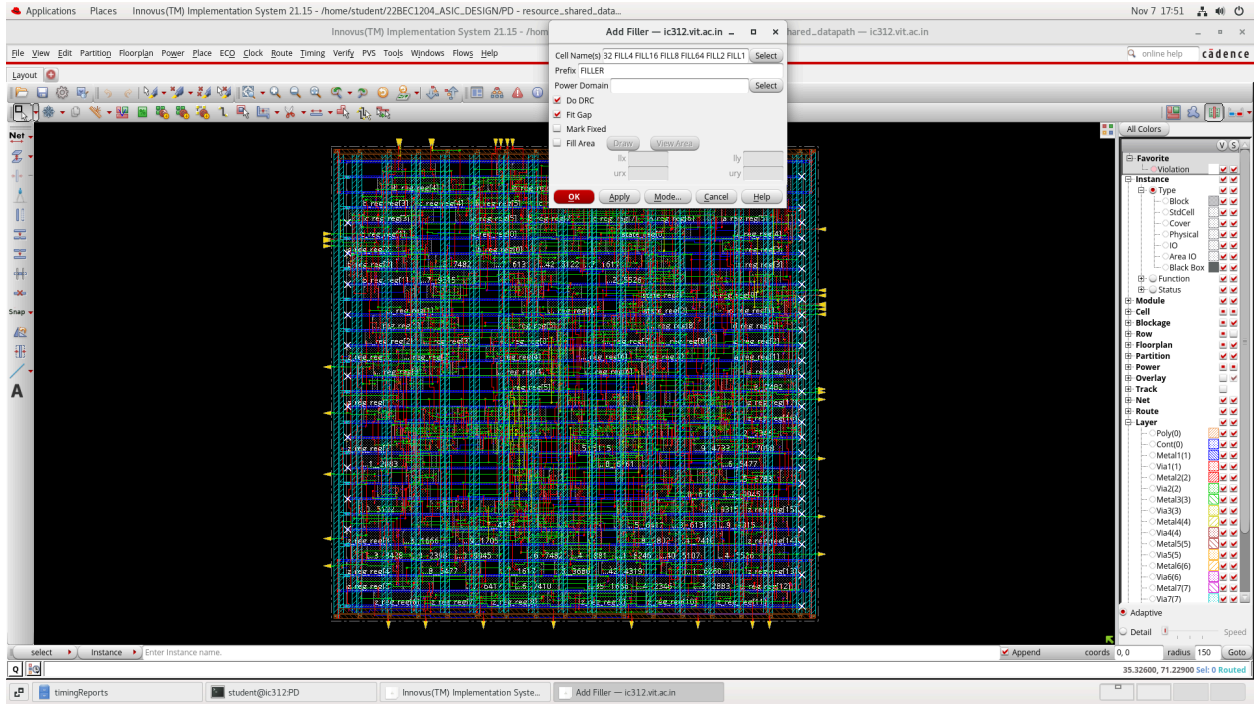
*** ccopt design #1 [finish] : cpu/real = 0:00:00.1/0:00:00.1 (0.9), totSession cpu/real = 0:02:03.5/0:21:58.6 (0.1), mem = 2014.2M
% End ccopt design (date=11/07 17:49:12, total cpu=0:00:00.1, real=0:00:00.0, peak res=1878.0M, current mem=1878.0M)

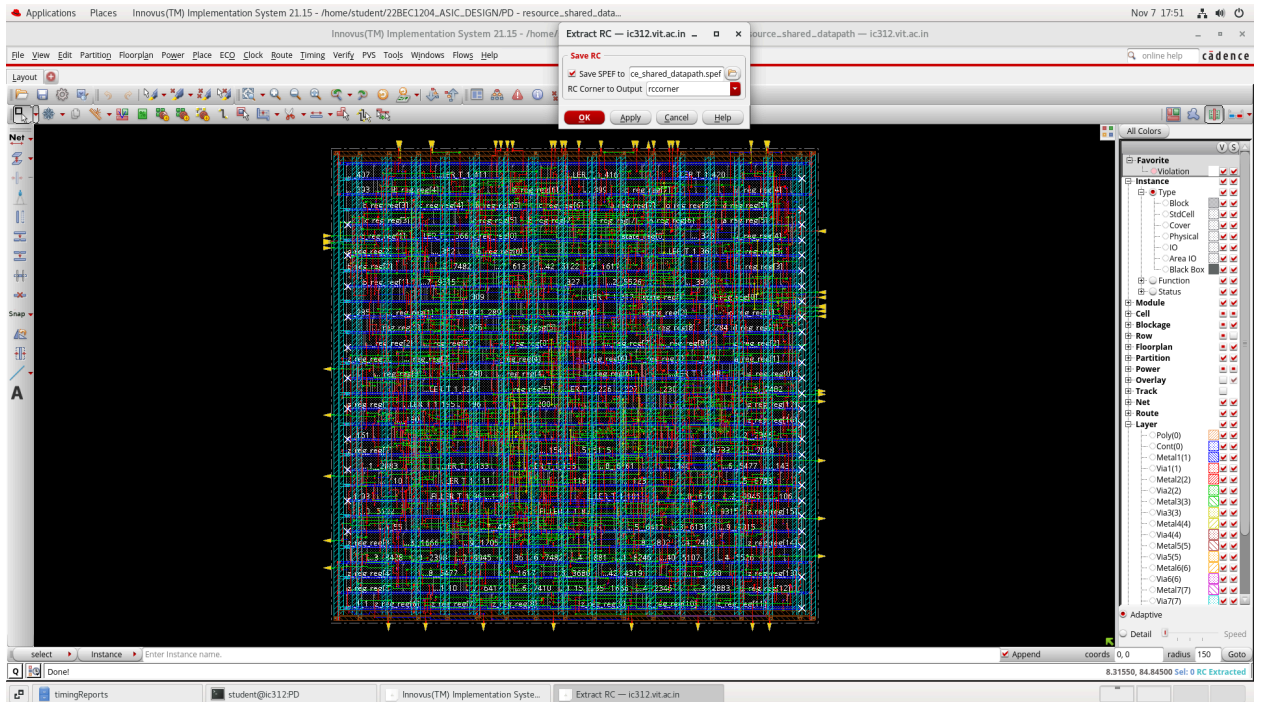
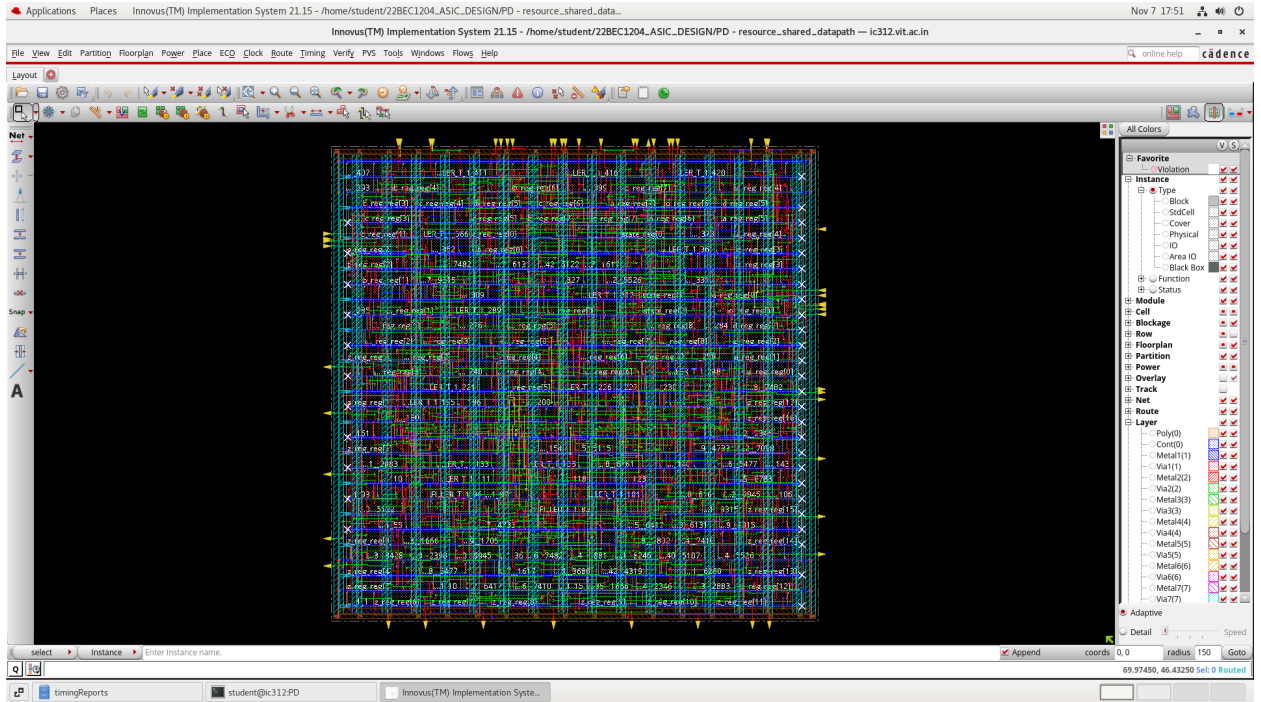
Innovus 3> saveDesign
DBS/ Default.globals Default.view ccopt.spec innovus.cmd innovus.cmd1 innovus.cmd2 innovus.log innovus.log1 innovus.log2 innovus.logv innovus.logv1
innovus.logv2 pipeline_cpu pipeline_cpu.dat/ pipeline_cpu.spf pipeline_cpu.v pvsUI_ipvs.log register_file register_file.dat/ register_file.spf register_file.v register_file_PD streamOut.map
timingReports/
Innovus 3> saveDesign DBS/cts.enc2
% Begin save design ... (date=11/07 17:50:12, mem=1883.6M)
% Begin Save ccopt configuration ... (date=11/07 17:50:12, mem=1883.6M)
% End Save ccopt configuration ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1884.1M, current mem=1884.1M)
% Begin Save netlist data ... (date=11/07 17:50:12, mem=1884.1M)
Writing Binary DB to DBS/cts.enc2.dat/resource_shared_datapath.v.bin in single-threaded mode...
% End Save netlist data ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1884.5M, current mem=1884.5M)
Saving symbol-table file ...
Saving congestion map file DBS/cts.enc2.dat/resource_shared_datapath.route.congmap.gz ...
% Begin Save AAE data ... (date=11/07 17:50:12, mem=1884.8M)
Saving AAE Data ...
% End Save AAE data ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1884.8M, current mem=1884.8M)
Saving preference file DBS/cts.enc2.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=11/07 17:50:12, mem=1886.0M)
Saving floorplan file ...
% End Save floorplan data ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1886.7M, current mem=1886.7M)
Saving Drc markers ...
... 27 markers are saved ...
... 0 geometry drc markers are saved ...
... 0 antenna drc markers are saved ...
% Begin Save placement data ... (date=11/07 17:50:12, mem=1886.7M)
% End Save placement data ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1887.0M, current mem=1887.0M)
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=11/07 17:50:12, total cpu=0:00:00.0, real=0:00:00.0, peak res=1887.0M, current mem=1887.0M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:01.0 mem=2017.0M) ***
% End Save routing data ... (date=11/07 17:50:13, total cpu=0:00:00.0, real=0:00:01.0, peak res=1887.3M, current mem=1887.3M)
Saving property file DBS/cts.enc2.dat/resource_shared_datapath.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=2020.0M) ***
% Begin Save power constraints data ... (date=11/07 17:50:13, mem=1888.0M)
% End Save power constraints data ... (date=11/07 17:50:13, total cpu=0:00:00.0, real=0:00:00.0, peak res=1888.0M, current mem=1888.0M)
Generated self-contained design cts.enc2.dat
% End save design ... (date=11/07 17:50:13, total cpu=0:00:00.2, real=0:00:01.0, peak res=1891.8M, current mem=1891.8M)
*** Message Summary: 0 warning(s), 0 error(s)

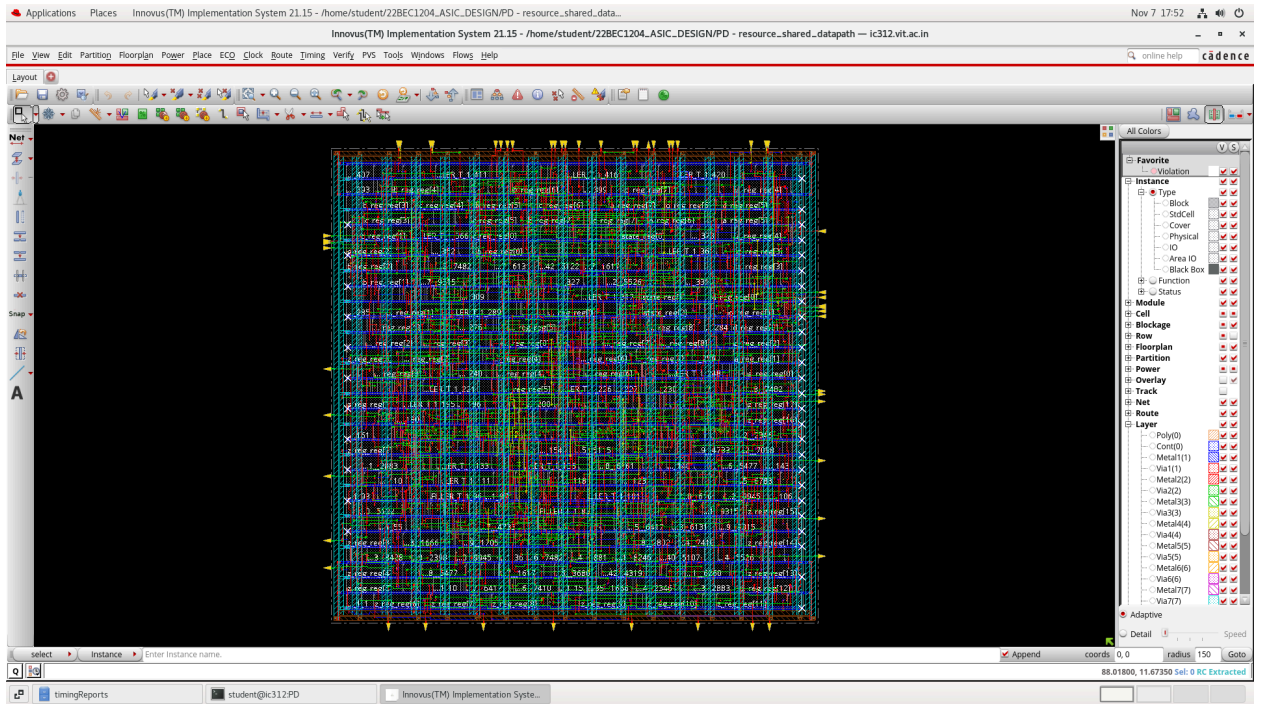
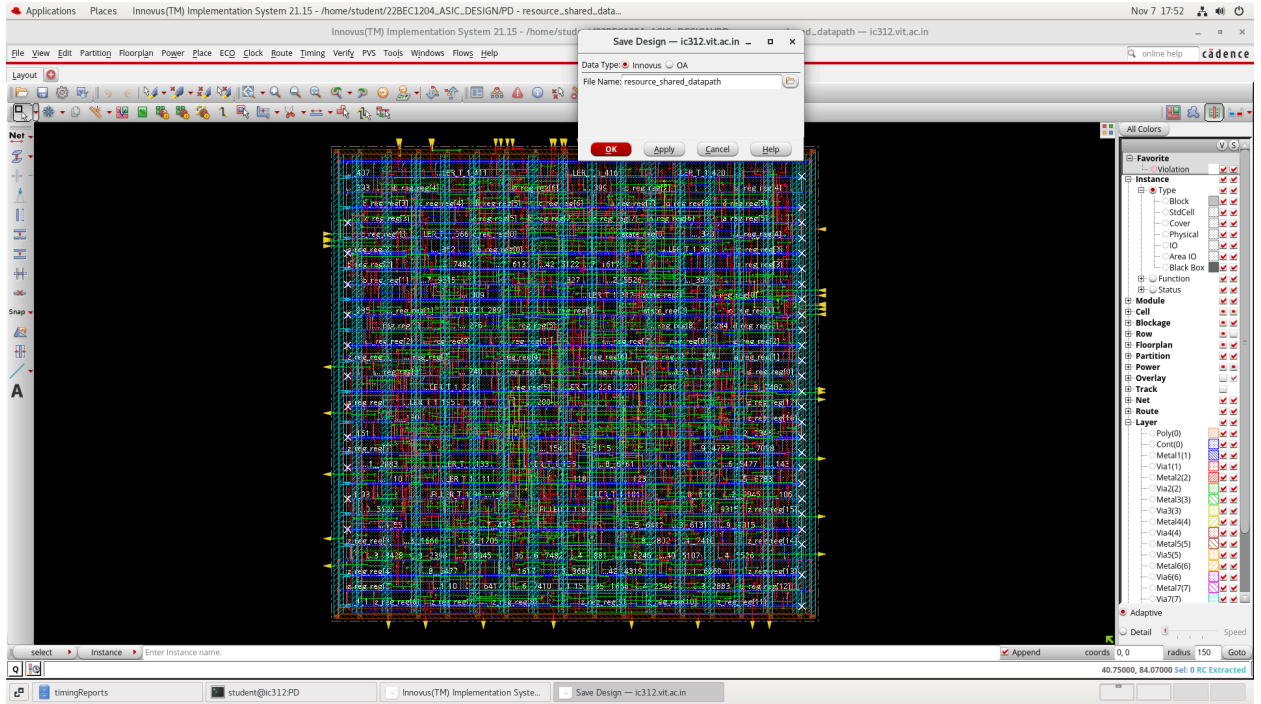
Innovus 4> 
```

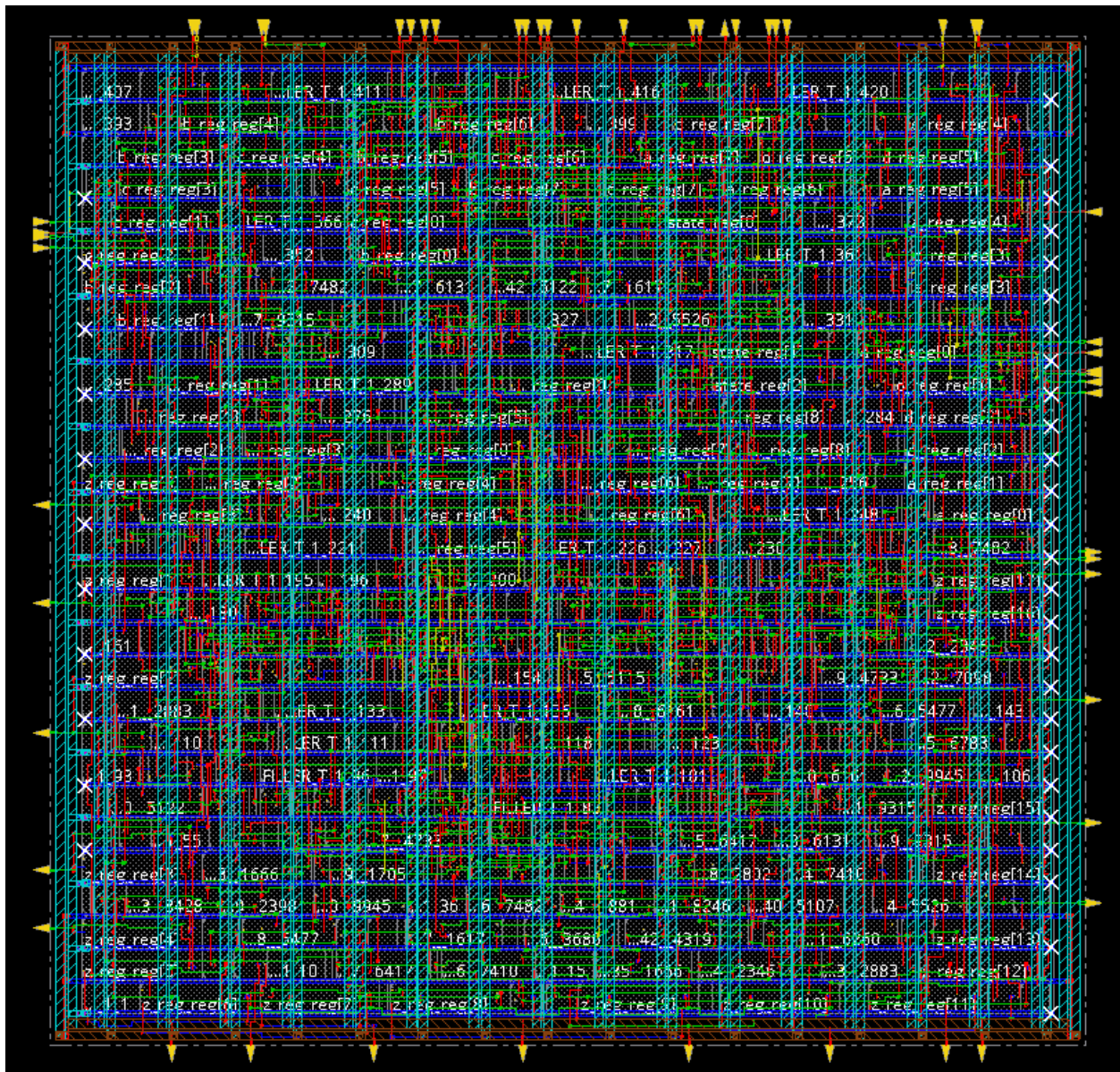


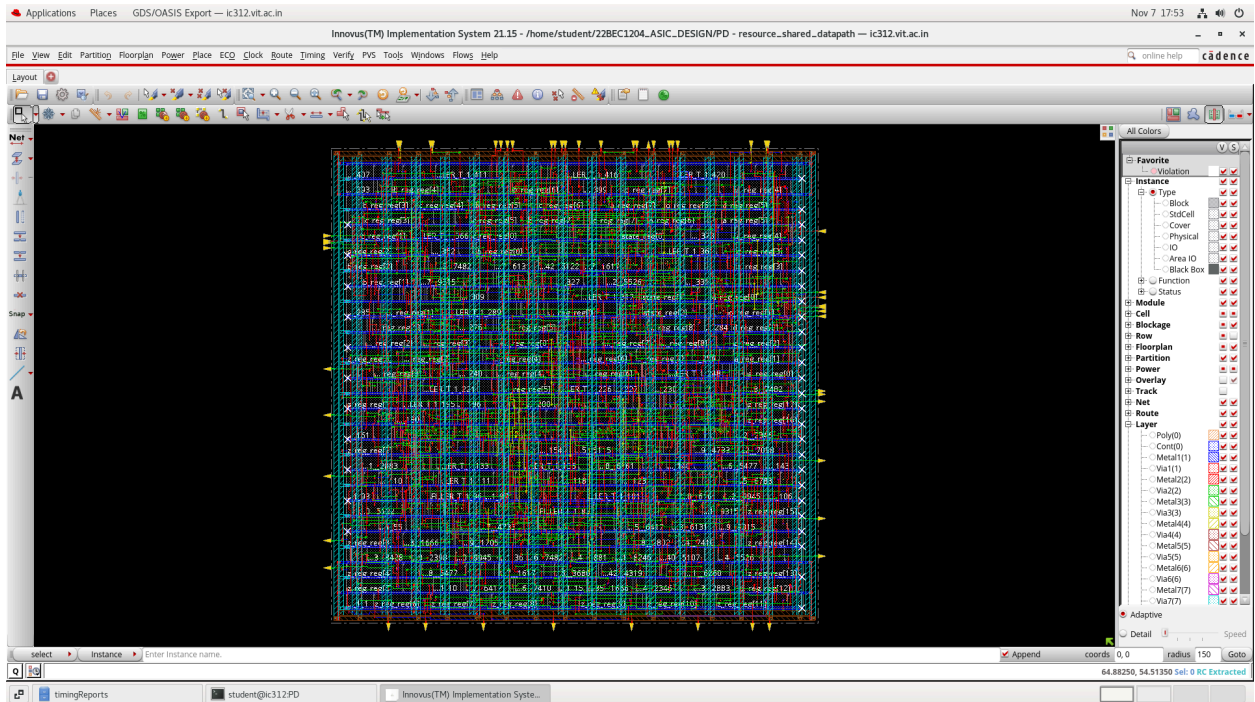
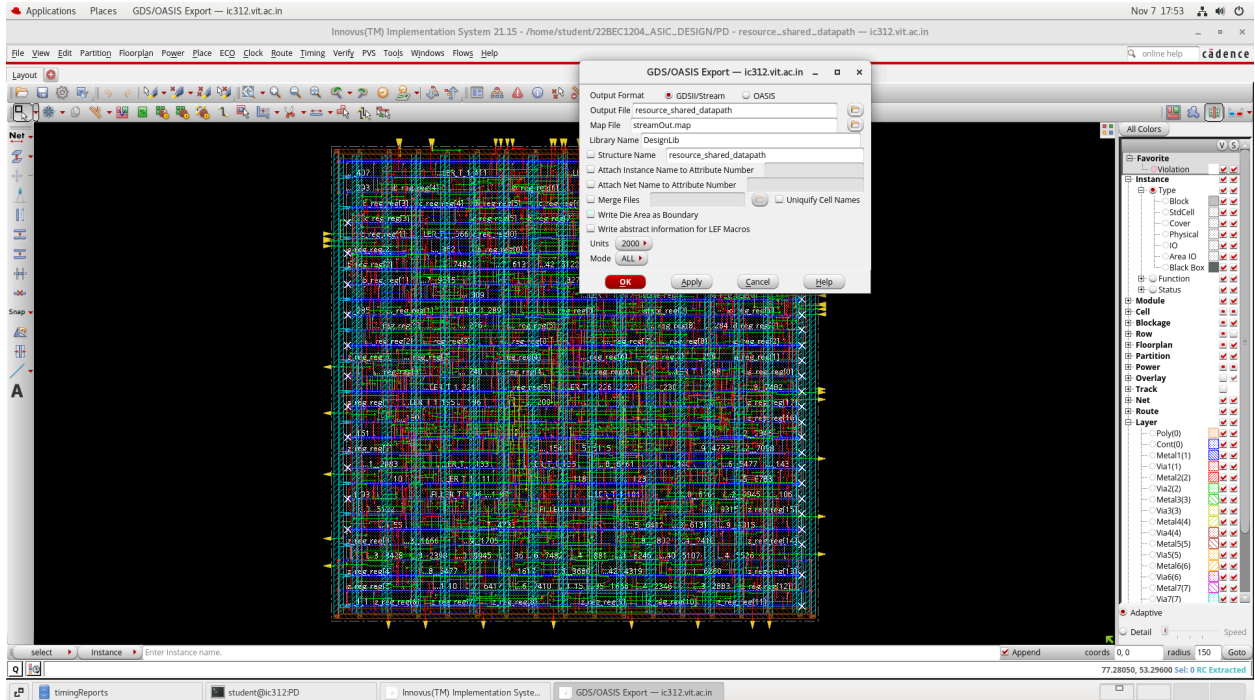


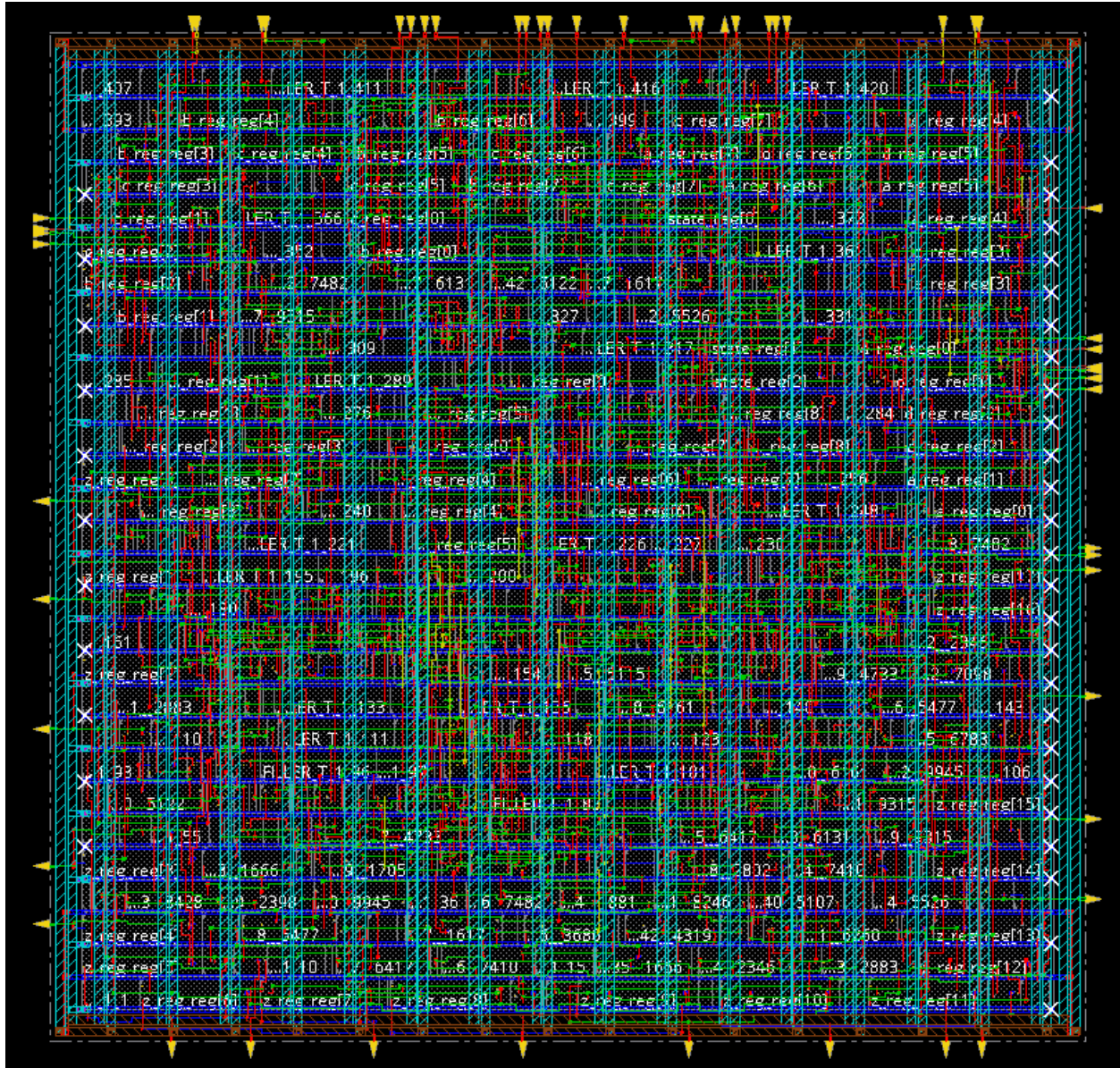












7. Inference

A multi-cycle datapath for $Z=(A+B)*(C-D)$ was successfully designed, verified, and implemented. The core objective, **resource sharing**, was achieved by using an FSM to schedule two operations (one add, one subtract) on a single ALU block over two clock cycles. This design trade-off reduces chip area (by eliminating a second ALU) at the cost of performance (5 cycles vs. 1 cycle).

The design passed the full RTL-to-GDSII flow, including DFT insertion and ATPG, resulting in a final, testable GDSII layout.